

Lucas Gagnon

Contact

USC Department of Mathematics
KAP Building, 3620 South Vermont Ave.
Los Angeles, CA 90089

Email: lgagnon@usc.edu
Web: <https://lucasgagnon.github.io>

Education

- Aug. 2022 Ph.D. in Mathematics, University of Colorado Boulder
“Advances in the combinatorics of the unipotent upper triangular groups,” sup. Nat Thiem
- 2019 M.A. in Mathematics, University of Colorado Boulder
- 2016 B.A. in Mathematics, Macalester College
Attended Budapest Semester in Mathematics Fall 2014

Academic Appointments

- 2025- RTPC Assistant Professor, University of Southern California
Mentored by Sami Assaf
- 2022-2025 Postdoctoral Visitor, York University
Mentored by Nantel Bergeron

Papers

Conventions: authors are always listed alphabetically by last name, and authors who were undergraduate (resp. grad.) students at the time of research are distinguished with an * (resp. **).

Published and forthcoming articles

- N. Bergeron, L. Gagnon, P. Nadeau, H. Spink, and V. Tewari. *The quasisymmetric flag variety: a toric complex on noncrossing partitions*. Forum of Mathematics, Pi 14 (2026).
- N. Andrews*, L. Gagnon, F. Gelinas**, E. Schlums*, and M. Zabrocki. *When are Hopf algebras determined by integer sequences?* Journal of Pure and Applied Algebra 230-9 (2026).
- N. Bergeron and L. Gagnon. *The excedance quotient of the Bruhat order, Quasisymmetric Varieties and Temperley–Lieb algebras*. Journal of the London Mathematical Society 110-4 (2024).
- L. Gagnon. *A unipotent realization of the chromatic quasisymmetric function*. Algebra & Number Theory 18-9 (2024), 1737–1766.
- L. Gagnon. *A $\mathrm{GL}(\mathbb{F}_q)$ -compatible Hopf algebra of unitriangular class functions*. Journal of Algebra 632 (2023), 426–461.
- L. Gagnon. *The combinatorics of normal subgroups in the unipotent upper triangular group*. Combinatorial Theory 1 (2021), Paper No. 15.

Refereed conference proceedings

- N. Bergeron, L. Gagnon, H. Spink, and V. Tewari. Extended Abstract: *Introducing the Coxeter Flag Variety*. Sémin. Loth. Comb. 96B: FPSAC Proceedings (2026).

N. Bergeron and L. Gagnon. Extended Abstract: *Excedance quotients, Quasisymmetric Varieties, and Temperley–Lieb algebras*. Sémin. Loth. Comb. 91B: FPSAC Proceedings (2024), Paper No. 67.

L. Gagnon. Extended Abstract: *A unipotent realization of the chromatic quasisymmetric function*. Sémin. Loth. Comb. 89B: FPSAC Proceedings (2023), Paper No. 72.

Preprints

L. Colmenarejo, L. Gagnon, and A. Ram. *Formulas for Koornwinder polynomials*. arXiv:2608.02810.

N. Bergeron, L. Gagnon, H. Spink, and V. Tewari. *The Quasisymmetric Grassmannian*. arXiv:2604.24903.

N. Bergeron, L. Gagnon, H. Spink, and V. Tewari. *The Coxeter flag variety*. arXiv:2601.23111.

N. Bergeron, L. Gagnon, P. Nadeau, H. Spink, and V. Tewari. *Equivariant quasisymmetry and noncrossing partitions*. arXiv:2504.15234.

In preparation

L. Gagnon, A. Woo, and Z. Zhuang^{**}. *Components of subregular Δ -Hessenberg varieties*. In preparation.

J. Beatty^{*}, L. Gagnon, P. Valdimiroff^{*}, J. Wen^{**}. *Puzzle rules for multiplying forest polynomials*. In preparation.

N. Bergeron, L. Gagnon, H. Spink, V. Tewari. *The peak quasisymmetric flag variety I: Schubert calculus*. In preparation.

L. Colmenarejo, L. Gagnon, N. Thieme, A. Ram. *Comparing Springer representations of Cherednik algebras*. In preparation.

Grants, Fellowships, and Awards

AMS-Simons Travel Grant, 2026-2028.

CU Boulder W. J. Thron Mathematics Fellowship, 2021.

CU Boulder Graduate School Summer Fellowship, 2020.

Macalester College Konhauser Award for Mathematical Achievement, 2016.

Invited Presentations

Upcoming

Representation and Lie Theory Seminar, University of California Riverside, October 20, 2026.

Special session on “Algebraic and Geometric Combinatorics,” AMS Western Sectional Meeting, Arizona State University, November 2026.

Scientific session on “Algebraic combinatorics: longstanding questions in symmetric functions,” CMS Winter Meeting, Montreal, Dec. 11-14, 2026.

Special session on “Crystals, Schubert Calculus, and Beyond,” Joint Math. Meetings, Chicago, Jan. 12-15 2027.

Special session on “Algebraic and Geometric Combinatorics,” Joint Math. Meetings, Chicago, Jan. 12-15 2027.

Past

Poster presentation, FPSAC (38th international conference on formal power series and algebraic combinatorics), University of Washington, July 14 2026.

Minisymposium on new perspectives in algebraic combinatorics, SIAM Conference on Discrete Mathematics, UC San Diego, June 22 2026.

Representation theory seminar, University of Melbourne, June 4, 2026.

Algebraic geometry & combinatorics seminar, Washington University in St. Louis, Nov. 11, 2025.

Combinatorics Seminar, UCLA, Nov. 6, 2025.

TACO (Topology, Algebra, Combinatorics, Operators) Seminar, Loyola University, Oct. 8, 2025.

Combinatorics and Geometry Seminar, University of Washington, April 9, 2025.

CAGE (Combinatorics, Algebra, and Geometry) Seminar, Philadelphia, Feb. 13 2025.

Workshop on non-crossing partitions, Erwin Schrödinger Institute, Feb. 17-21, 2025.

Combinatorics seminar, University of Southern California, Dec. 9 2024.

Bay Area Discrete Math Day, UC Berkeley, Dec. 7, 2024.

Algebra & discrete mathematics seminar, UC Davis, Dec. 6 2024.

Combinatorics seminar, Stanford University, Dec. 5 2024.

Séminaire du LACIM, Université du Québec à Montréal, Nov. 22, 2024.

Discrete geometry and combinatorics seminar, Cornell University, Nov. 11 2024.

Poster presentation, FPSAC (36th international conference on formal power series and algebraic combinatorics), Ruhr-Universität Bochum, July 22-26, 2024.

Special session on groups and representations, AMS Spring Western Sectional Meeting, San Francisco State University, May 4-5, 2024.

Scientific session on algebraic and enumerative combinatorics, CMS Winter Meetings, Montreal, Dec. 3, 2023.

Special session on combinatorial and categorical techniques in representation theory, AMS Fall Eastern Sectional Meeting, the University at Buffalo, Sept. 9, 2023.

Poster presentation, FPSAC (35th international conference on formal power series and algebraic combinatorics), UC Davis, July 20, 2023.

Algebra and geometry seminar, Queen's University, April 17, 2023.

Algebraic and enumerative combinatorics seminar, University of Waterloo, Mar. 23, 2023.

Séminaire du LACIM, Université du Québec à Montréal, Mar. 10, 2023.

Discrete math seminar, University of British Columbia (Online), Feb. 14, 2023.

Combinatorial algebra meets algebraic combinatorics conference (CAAC), University of Waterloo, Jan. 20, 2023.

Combinatorics seminar, Washington University in St. Louis, Dec. 8, 2022

Scientific session on algebraic combinatorics and representation theory, CMS Winter Meetings, Toronto, Dec. 3, 2022

Special session on finite groups, their representations, and related structures, AMS Spring Western Sectional Meeting, Online, May 14, 2022.

Rocky mountain algebraic combinatorics seminar, Colorado State University, May 6, 2022.

Discrete mathematics seminar, University of British Columbia (Online), Mar. 29, 2022

Algebra seminar, Binghamton University (Online), May 11, 2021.

Special session on group representations and related topics, AMS Fall Eastern Sectional Meeting, Online, Oct. 3, 2020.

Undergraduate Researchers Mentored

USC MSRP Summer 2026	<i>Forest polynomials and a Thompson analogue of the Fomin–Kirillov algebra.</i> Mentees Justin Beatty and Parker Vladimiroff, and graduate assistant Jiayi Wen.
Summer Project Summer 2024	<i>Using machine learning to find combinatorial bijections.</i> Mentees Inderpreet Ganda and Eniola Ogunjimi and graduate assistant Felix Gelin. Jointly advised with M. Zabrocki.
Summer Project Summer 2024	<i>Catalan Hopf subalgebras of NCSym.</i> Mentees Nicholas Andrews and Eric Schlums and graduate assistant Felix Gelin. Jointly advised with M. Zabrocki.
Honors Thesis AYs 2020–2022	<i>Motivating a q-version of the embedding between the categories Heis and $\text{Par}(t)$.</i> Mentee Theodore Gonzales. Jointly advised with T. Xu.
REU Mentor Summer 2020	<i>Rewiring matrices using combinatorial tools.</i> Mentees Jacob Gaiter, Juliet Hawkes, Tom Kauffman, Adam Nelson. Supervised by N. Thiem at CU Boulder.

Teaching Credentials

- 2026 Certificate in College Teaching, University of Colorado Boulder
- 2023 Instructional Skills Intensive Workshop, York University

Courses Taught

University of Southern California

- Applied Combinatorics (Spring 2026, Fall 2026)
- Precalculus for Business and Economics (Fall 2025)

York University

- Abstract Linear Algebra (Spring 2025)
- Mathematics for Life Sciences I (Spring 2024)
- Conjecture, Theorem, and Proof (Spring 2023)

University of Colorado Boulder

- Assistant Course Coordinator for Multivariable Calculus (Fall 2021, Spring 2022)
- SASC Calculus Coseminar (Spring 2020, Fall 2020, Spring 2021)
- Differential Calculus (Instructor of Record: Fall 2019, Recitation TA: Fall 2017)
- Integral Calculus (Instructor of Record: Fall 2018, Spring 2019, Recitation TA: Spring 2019)

Service

Departmental Service

2026–present	Faculty advisor, USC Go club.
2025–present	Organizer, USC Combinatorics Seminar
2022–2025	Organizer, York Applied Algebra Seminar
April 19 2022	Organizer, Workshop on Campus and Departmental Culture, CU Mathematics
2021–2022	Member, CU Mathematics Diversity Committee
2020–2021	Peer Mentor, CU Mathematics Graduate Mentoring Program
2019–2021	Graduate Representative, CU Mathematics Undergraduate Committee
2018–2020	Organizer, Graduate Student Seminar, CU Mathematics

Professional Service

June 7-8 2023	Organizer, “Many perspectives on Hopf Algebras in Combinatorics” Canadian Discrete and Algorithmic Mathematics Conference contributed minisymposium.
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Reviewer for: Advances in Mathematics; Algebraic Combinatorics; Annals of Combinatorics; Communications in Algebra; Communications of the American Mathematical Society; Electronic Journal of Combinatorics; European Journal of Combinatorics; Forum of Mathematics, Sigma; MathSciNet; Journal of the London Mathematical Society, Transactions of the American Mathematical Society.